

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	XXXXXXXX
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project predicts the Human Development Index (HDI) using a Linear Regression machine learning model. It provides a simple web-based application for predicting HDI from key socioeconomic indicators. The system delivers fast, accurate predictions and classifies countries into Low, Medium, High, or Very High human development.

Project Overview	
Objective	Predict a country's Human Development Index (HDI) score and development category using Linear Regression based on Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita.
Scope	Develop a complete machine learning pipeline including data preprocessing, exploratory data analysis (EDA), model training, evaluation, and deployment through a Flask web application.
Problem Statement	
Description	There is no simple and interactive system that allows users to enter development indicators and instantly predict a country's HDI without technical knowledge.
Impact	The proposed solution provides a user-friendly web application that delivers quick and accurate HDI predictions using a pre-trained machine learning model.
Proposed Solution	
Approach	Approach: Train a Scikit-learn Linear Regression model on the Kaggle Human Development Index dataset after preprocessing and handling missing values, then deploy the trained model using Flask for real-time HDI prediction.
Key Features	• Linear Regression HDI score prediction (0–1) • Input validation on all fields • REST API at /api/predict • Jupyter Notebook with full EDA (heatmap, scatter, pairplot, residual)

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU / GPU	Standard laptop CPU (no GPU needed for Linear Regression)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv